

Timeline Check in

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Ideas for analysis

Questions

- How does the percent cover of native grasses vary over time compared to the percent cover of exotic vegetation on the NCOS Mesa Perennial Grasslands?
 - How does intentional disturbance influence this relationship?
 - How do key individual species, including *Stipa pulchra*, *Medicago polymorpha*, *Melilotus indicus*, and *Festuca perennis*, vary in percent cover before and after intentional disturbance?

Statistical analysis

- statistical analysis to address at least one question with at least one analysis coded and run

Background

We assess whether the 2023 cultural burn is meeting its restoration goals of promoting native grassland species. This study may help to inform future restoration management decisions regarding intentional disturbance at NCOS and could have implications which extend to the management or ecological understanding of other coastal grassland systems. These questions are relevant because Santa Barbara coastal grasslands are fire adapted (Saglimbeni, 2024, pp. 1-2), but are now heavily affected by invasive species. Studying the NCOS burn helps show whether cultural burning can support native species like *Stipa pulchra* while reducing nonnative grassland invaders.

Visualizations

Plot 1: Native and exotic percent cover over time

Plot 2: Sweet and bur clover abundance over time

Plot 3: Change in mean percent cover of key species after disturbance

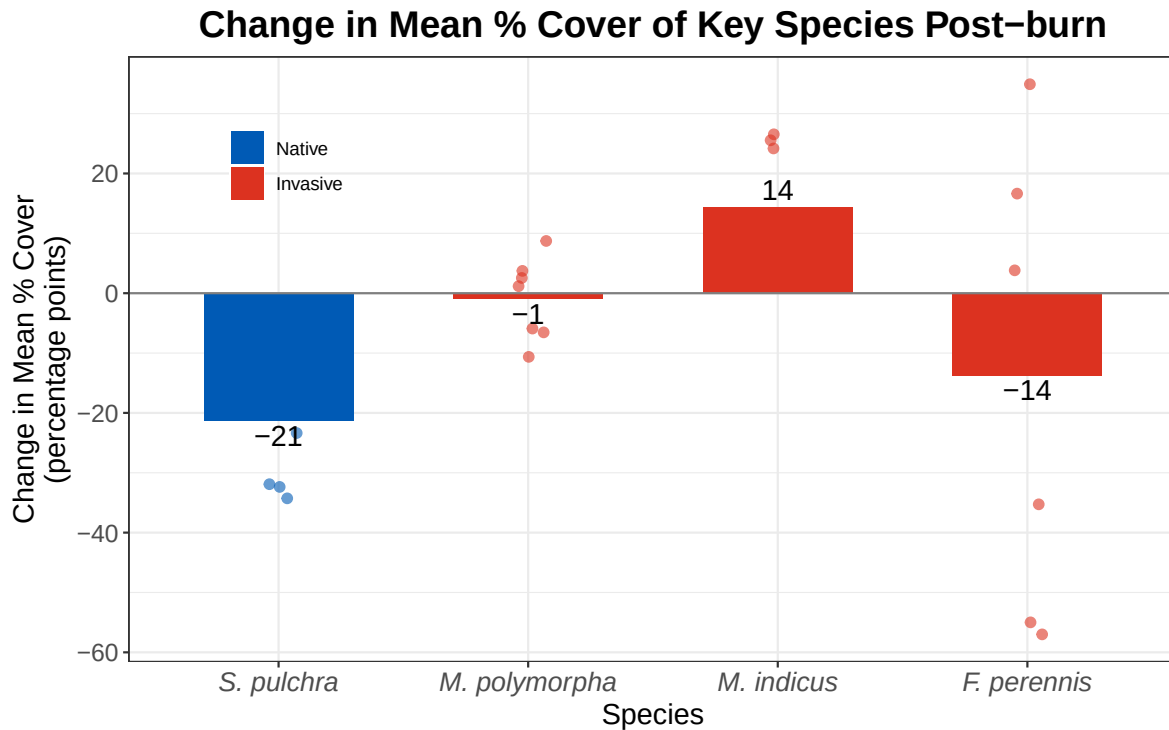


Figure 1: Change in mean percent cover of four key grassland species on the NCOS Mesa from 2023, pre-burn, to 2024, one year post-burn. Bars represent the average percent-point change in cover across transects for each species, while individual points show transect-level percent-point changes; the gray horizontal line represents no change, with values above zero indicating increased cover and values below zero indicating decreased cover. Blue indicates the native species, *Stipa pulchra*, and red indicates invasive species: *Medicago polymorpha*, *Melilotus indicus*, and *Festuca perennis*. Overall, *S. pulchra* and *F. perennis* decreased after the burn, *M. indicus* increased, and *M. polymorpha* showed little average change. This figure connects to our main question by showing that intentional disturbance did not affect all native and invasive species in the same way.

Plot 4: Mean percent cover of native grasses after disturbance

Exploratory Visualization 1: Change in mean percent cover of key species after sheep grazing

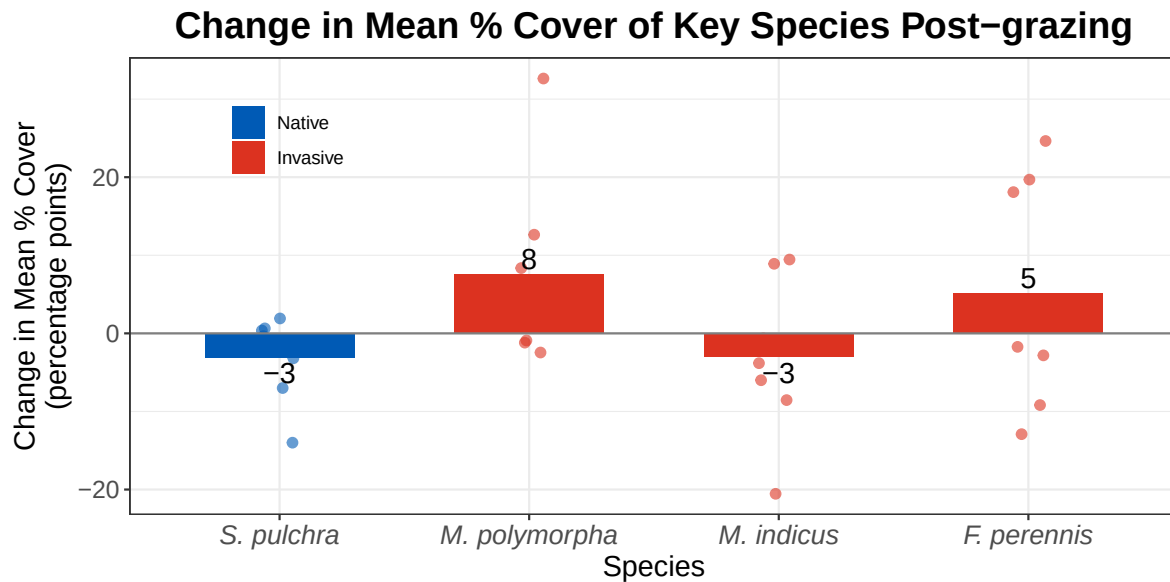


Figure 2: Change in mean percent cover of four key grassland species on the NCOS Mesa from 2024, pre-grazing, to 2025, post-grazing. Bars show the average percent-point change across transects, individual points show transect-level changes, and the gray horizontal line represents no change; blue indicates the native species *Stipa pulchra*, while red indicates invasive species. As an exploratory visualization, this figure suggests that *S. pulchra* changed only slightly after grazing, while invasive species showed mixed responses, with *M. polymorpha* and *F. perennis* increasing on average and *M. indicus* decreasing slightly.

Exploratory Visualization 2: Effect of annual precipitation on native and invasive percent cover

Plan for elective

- the general concept
- a description of the progress you have made
- a link, photo, or some visual of your elective